VORLDX

EMPLOYEE ATTENDANCE SYSTEM

Functional Requirements Specification
Architecture • Roles • Database • Workflows • Security • Acceptance Tests

Technology: React · React Native · Supabase

QR policy: 18-second rotation **WFH cutoff:** 2 hours before shift

Project specification prepared for implementation planning and requirement verification.

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Purpose

This document consolidates the original VorldX internship brief, the user-provided workflow diagrams, and the six decisions confirmed by the project owner. It defines expected behavior and a proposed technical design. Proposed implementation details are identified where the source brief does not prescribe a specific approach.

Key decisions

Area	Confirmed rule
Initial company Admin	Company signup creates the initial Admin automatically.
WFH approvals	HR approves employee requests; Admin approves HR requests.
WFH attendance	Approved request requires a single-use passcode to mark attendance.
Policies	Admin controls global policies; HR can view them.
Daily attendance	One check-in and one check-out per employee per day.
Feedback	Ratings out of 10 and written comments.
Office QR	QR session refresh every 18 seconds.
Stack	React website, React Native app, Supabase backend.

Contents

This document is organized to support review first and implementation afterward.

- 1. Project scope and objectives
- 2. System architecture
- 3. Roles and permissions
- 4. Functional requirements
- 5. Database schema
- 6. Authentication and onboarding flows
- 7. Attendance and 18-second QR workflow
- 8. Work From Home workflow
- 9. Feedback and performance
- 10. Reporting and statistics
- 11. Security and privacy
- 12. UI and navigation structure
- 13. Error handling and edge cases
- 14. Acceptance criteria and testing
- 15. Implementation roadmap
- 16. Project report and demo scenario
- 17. Decisions and assumptions register

How to read this specification

Confirmed requirement means explicitly stated by the project owner or source brief. **Proposed rule** means a recommended implementation detail that should be reviewed if it affects company policy. The system must use dummy data for demonstrations and must not use real employee personal information in demo seed records.

1. Project scope and objectives

The VorldX Employee Attendance System is a multi-company attendance platform. It contains a Company Portal for onboarding, an Admin/HR website, an employee mobile app, and a supporting office QR display. All interfaces use one Supabase project and share persistent records.

Objectives

- Provide controlled company, HR, and employee onboarding with role-based access.
- Record exact check-in/check-out timestamps and distinguish office attendance from WFH attendance.
- Rotate office QR sessions every 18 seconds and validate them on the backend.
- Enforce the two-hour WFH request deadline based on the user's assigned shift.
- Use the approval hierarchy: HR approves employee WFH; Admin approves HR WFH.
- Require a single-use passcode to mark attendance for an approved WFH request.
- Provide filters, accurate statistics, charts, feedback ratings, and written comments.
- Demonstrate all flows using clearly identified fictional sample data.

In scope

- Company signup and initial Admin creation; HR invitations; employee records and account activation.
- Email/password login, password recovery, and optional local biometric unlock in the mobile app.
- Employee profiles, attendance, WFH requests, feedback, and personal statistics.
- Admin/HR dashboards, employee/HR management, policies, QR display control, approvals, reporting, and audit records.

Out of scope for the initial version

- GPS or location-based tracking, Wi-Fi/Bluetooth proximity, face/fingerprint attendance verification, payroll, complex leave management, and advanced manager workflows.
- Biometric unlock is a device-local convenience feature; biometric templates are not stored by this application.

2. System architecture

The system uses React + Vite + Tailwind CSS for the website, React Native + Expo for the employee app, and Supabase for the backend platform. A separate Express server is not required.

Architecture overview

Company Portal	Company registration, initial Admin onboarding, login and recovery.
Admin/HR Website	Management, attendance, policies, WFH approvals, QR control, feedback and reports.
Employee App	Account activation, login, biometrics, QR scanner, WFH, attendance history and feedback.
QR Display	Authorized office screen that requests and renders short-lived QR sessions.
Supabase Auth	Authentication sessions, email confirmation and password recovery.
PostgreSQL + RLS	Persistent data, relational constraints and company/role/ownership-based access.
Edge Functions / SQL	Privileged business rules, QR issuance, attendance transactions, passcodes and approvals.

Frontend operations should use the Supabase client for authorized reads and simple operations. Privileged actions must be validated by Edge Functions or carefully secured database functions. The service-role key must never be shipped to browser or mobile code.

Recommended libraries

Area	Options
Website	React, Vite, Tailwind CSS, React Router, Recharts, React Hook Form, Zod
Mobile	React Native, Expo, expo-camera, expo-local-authentication
QR	qrcode (rendering); cryptographically secure token generation on server
Backend platform	Supabase Auth, PostgreSQL, RLS, Edge Functions, SQL functions, optional Realtime
Testing	Vitest/Jest as appropriate, Supabase local development, Postman/Bruno, end-to-end checks

3. Roles and permissions

Roles are company-scoped. A user may only access data for companies where they have an active membership. Role assignment is performed by trusted server-side code, not by the signup form.

Capability	Admin	HR	Employee
Company profile and global settings	Full access	View only	No
Create/deactivate HR accounts	Yes	No	No
Create/edit/deactivate employees	Yes	Yes	No
Edit global policies	Yes	No	No
View applicable policies	Yes	Yes	Relevant rules only
Approve employee WFH	No by default	Yes	No
Approve HR WFH	Yes	No	No
Generate WFH passcode	For HR request	For approved employee request	Use own passcode
Mark own attendance	Yes	Yes	Yes
View own attendance	Yes	Yes	Yes
View employee attendance	Company-wide	Authorized company scope	Own only
View HR attendance	Company-wide	Own only	No
Control QR displays	Yes	No by default	No
Review employee feedback	No by default	Yes	No
Review HR feedback	Yes	No	No
Read own feedback	Yes	Yes	Yes
View audit logs	Yes	Limited if authorized	No

Permission notes

- Admin has company-wide control, including HR account management and QR display pairing.
- HR can manage employee records and attendance within the authorized company scope, but cannot change global policies.
- HR approves employee WFH requests. Admin approves HR WFH requests. These are separate reviewer rules.
- Users cannot approve their own WFH request, change their own role, or edit their own feedback ratings.
- UI route guards improve usability; RLS and server-side authorization provide actual enforcement.

4. Functional requirements

4.1 Company portal and account management

- Company signup collects company name, company email, and initial Admin credentials. Company identifier is generated by the system.
- Company creation and initial Admin membership are performed through a trusted operation with failure-safe handling.
- HR signup is allowed only after Admin has created/invited the HR record. Employee activation requires a matching active employee record.
- Login uses email/password; password recovery uses Supabase Auth email reset links.
- Email confirmation, pending activation, and suspended membership states must be handled explicitly.

4.2 Employee app

- First-time setup: choose company, enter registered email and password, validate the pre-created employee record, activate account, and fetch employee details.
- Display name, employee ID, department, designation, joining date, shift and work type from the database.
- Offer local biometric unlock after successful authentication. It protects an existing session; it does not independently prove employment or mark attendance.
- Dashboard shows current attendance, exact timestamps, duration when complete, attendance type, WFH status, summaries, and shortcuts.
- QR scanner requests camera permission only when opened and sends the scanned token to the backend.
- Attendance history supports date, status, type, and time filters.
- Employee can submit WFH requests, see status/reason, and view feedback addressed to them.

4.3 Admin website

- Profile, company details, HR account creation/invitation/deactivation, HR attendance and global policy configuration.
- Review HR WFH requests, approve/reject, and authorize passcode generation for approved requests.
- Pair/activate/deactivate office QR displays and review display health/status.
- View/filter authorized attendance, perform audited corrections, and access company reports and HR feedback.

4.4 HR website

- Add/edit/deactivate employees, assign department, designation, joining date, work type and shift.
- View global attendance policies; do not edit them.
- Review employee WFH requests, approve/reject, and generate passcodes for approved requests.
- View/filter employee attendance, mark own attendance, view own attendance, and access reports.
- Create employee feedback with a review period, criterion ratings from 0 to 10, and written comments.
- View feedback and trends submitted by Admin about the HR user.

4.5 QR display

- Display must be paired/authorized by Admin before it can request valid QR sessions.
- The server issues a cryptographically random token, stores its hash and expiry, and returns a payload for rendering.
- The display requests a replacement every 18 seconds. Expiry is enforced on the server, not by a client countdown.
- When offline or unable to refresh, the display must show an error/expired state and must not present the old QR as valid.

5. Database schema

The following is a logical schema proposal. Use UUID primary keys, foreign keys, indexes, timestamps, constraints, and company-scoped uniqueness. Exact SQL types and normalization can be refined during migration design.

Table	Principal columns	Purpose
companies	id, name, slug, timezone, is_active, created_at	Organization and company settings
profiles	id (= auth.users.id), full_name, avatar_url, created_at, updated_at	User-facing profile
company_memberships	id, company_id, user_id, role, status, created_at	Company-scoped role and membership
employees	id, company_id, user_id?, employee_code, full_name, email, department, designation, joining_date, shift_id, work_type, is_active	HR-created employment record; account link may be null until activation
shifts	id, company_id, name, start_time, end_time, grace_period_minutes, working_days, is_active	Assigned work schedules
attendance_policies	id, company_id, wfh_enabled, wfh_eligibility_rules, request_cutoff_minutes, updated_by, updated_at	Admin-managed global policy
attendance_records	id, company_id, employee_id, work_date, attendance_type, check_in_at, check_out_at, status, source, created_at, updated_at	One daily attendance record
wfh_requests	id, company_id, employee_id, request_date, reason, status, reviewed_by, reviewed_at, review_notes, created_at, updated_at	WFH request and approval state
wfh_passcodes	id, company_id, wfh_request_id, employee_id, work_date, passcode_hash, expires_at, used_at, generated_by, created_at	One-time passcode metadata
qr_displays	id, company_id, display_name, office_name, is_active, paired_at, last_seen_at	Registered office display
qr_sessions	id, company_id, display_id, token_hash, issued_at, expires_at, revoked_at	Short-lived QR session metadata
feedback_reviews	id, company_id, subject_employee_id, reviewer_user_id, period_start, period_end, comments, status, created_at, updated_at	Review header and comments
feedback_ratings	id, review_id, criterion, score, comments?	Individual rating per criterion; score 0–10
audit_logs	id, company_id, actor_user_id, action, entity_type, entity_id, old_values, new_values, created_at	Sensitive-action history

Key constraints and indexes

- UNIQUE(company_id, employee_code) on employees; enforce normalized email uniqueness within the intended company scope.
- UNIQUE(company_id, user_id) on company_memberships.
- UNIQUE(company_id, employee_id, work_date) on attendance_records for the one-record-per-day model.
- CHECK(feedback_ratings.score BETWEEN 0 AND 10).
- Index attendance by company/date, employee/date, status and attendance type; index pending WFH requests by company/status/request date.
- Passcode/QR hashes should be unique where appropriate. Use transactions/row locks to enforce single-use consumption.
- Keep raw tokens and plaintext passcodes out of logs. Use audit logs for changes, not as a substitute for the current record.

6. Authentication and onboarding workflows

6.1 Company signup

User enters company details and initial Admin credentials. A trusted operation validates the request, creates the Auth account, creates the company, and creates the initial Admin membership. The role is assigned by server-side code. If email confirmation is required, the user remains pending until confirmation succeeds.

6.2 HR activation

- Admin creates the HR record and sends an authorized invitation.
- HR follows the activation flow and proves control of the invited email.
- The system links the Auth user to the HR employment record and assigns the HR role.
- HR logs in and can access HR routes and authorized company records.

6.3 Employee activation

- HR creates the employee record before signup.
- Employee selects company and enters the registered email/password.
- Server checks the company, active employee record, invitation/activation state and email ownership.
- Account is linked to the existing employee record; the app fetches assigned employee details.
- Employee enters the dashboard and may enable local biometric unlock.

6.4 Login and recovery

- Supabase Auth validates credentials and issues the session.
- App/website obtains the user's active company membership and role from trusted records.
- Route guards direct the user to the correct dashboard.
- Forgot password sends a reset link via Supabase Auth.
- Suspended users and inactive employees are blocked from protected attendance operations.

6.5 Biometric unlock

Use device-local biometric authentication (for example, Expo Local Authentication) to unlock a protected session or credential. Do not upload or store raw biometric templates. Provide secure fallback reauthentication. Biometrics must not bypass server authorization or replace QR/passcode validation.

7. Attendance and 18-second QR workflow

7.1 Daily attendance state

Current state	Action	Expected result
No record	Valid office QR	Create daily record with check-in timestamp
Checked in	Valid office QR	Set check-out timestamp on existing record
Checked out	Scan again	Reject; do not create another daily record
No record	Valid approved WFH passcode	Create WFH record with required timestamps/policy
Existing office/WFH record	Attempt conflicting attendance type	Reject; never silently overwrite
Expired/invalid QR	Scan	Reject and request a fresh QR

7.2 QR lifecycle

- Admin pairs an office display. The display receives permission to request sessions only for its assigned company/display.
- Edge Function creates a cryptographically random token, stores a token hash and issued/expiry timestamps, and returns the payload.
- Display renders the QR and a countdown. It requests a replacement every 18 seconds.
- Mobile app submits the scanned token with the authenticated session.
- Backend checks token hash, expiry, display status, company membership, active employee, and daily attendance state.
- A transaction records the attendance change and makes repeated requests idempotent where appropriate.

7.3 Timestamp and late rules

The server records timestamps. The work date is derived in the company's configured timezone. Late status is determined using the assigned shift start and grace period. For example, a 09:00 shift with a 10-minute grace period is on time through 09:10 and late after that point; the precise boundary must be defined consistently in the policy.

7.4 Check-out and incomplete records

Only a checked-in record can be checked out. If no check-out is recorded, show In progress during the shift or Incomplete after the applicable day/shift closes. Never fabricate an end time or duration.

7.5 QR security caveat

An 18-second lifetime reduces the QR reuse window but does not prevent screenshots being shared within that window. Server-side expiry, random tokens, rate limiting, authenticated users, display pairing, and atomic attendance updates are required. Avoid relying on client clocks.

8. Work From Home workflow

8.1 Request deadline

All employees and HR users must submit a request at least 120 minutes before the start of their assigned shift. Deadline = shift start – 120 minutes. A 09:00 shift therefore has a 07:00 deadline. The backend evaluates the deadline using trusted time and the company timezone.

8.2 Approval hierarchy

Requester	Reviewer	Decision authority
Employee	HR	HR approves/rejects employee request
HR	Admin	Admin approves/rejects HR request

8.3 Request states

pending → approved or rejected; cancellation may be allowed only before approval/attendance, according to policy. Review action stores reviewer identity, decision time, and optional notes. A user cannot review their own request.

8.4 Passcode and attendance

- Passcode generation is available only after an approved request and to an authorized reviewer role.
- Passcode is random, bound to request + employee + date, expires, is single-use, and is stored as a hash.
- On entry, backend validates the authenticated user, approved request, date, expiry, unused status, eligibility and absence of conflicting attendance.
- Within one transaction, create the WFH attendance record and mark the passcode used.
- Rejected, pending, expired or already-used passcodes never create attendance.
- If a record already exists for that day, reject the operation instead of overwriting office attendance.

8.5 WFH timestamps

The passcode authorizes WFH attendance type; it does not itself supply a fabricated check-in/out time. To meet the exact-timestamp requirement consistently, the user should use an explicit WFH check-in and check-out flow after validating the approved passcode. Whether the passcode is required once per day or at the start of the WFH session should be implemented as one consistent policy.

9. Feedback and performance

9.1 Review flows

Reviewer	Subject	Capabilities
HR	Employee	Select review period, rate criteria 0–10, enter comments, save review
Admin	HR	Select review period, rate criteria 0–10, enter comments, save review

9.2 Suggested criteria

- Punctuality
- Attendance and consistency
- Teamwork and collaboration
- Adaptability
- Accountability

Criteria should be configurable only if the product needs that flexibility. Each rating must be an integer from 0 to 10, linked to a review and subject. A review contains the period start/end, reviewer, comments, and submission timestamps.

9.3 Employee/HR feedback view

- Show review period, reviewer, criterion ratings and written comments.
- Display criterion-wise charts and historical trends across review periods.
- Allow filters by date range where the user has access to the records.
- Subjects cannot edit or delete submitted ratings. Corrections should be controlled and audited.

9.4 Interpretation

Attendance metrics and qualitative feedback ratings are separate measures. Do not combine them into a single performance score unless a defined and approved formula is added and disclosed.

10. Reporting and statistics

Employee dashboard metrics

- Present, late and qualifying WFH days.
- Attendance percentage using the documented eligible-working-day denominator.
- Today's check-in/check-out times and completed duration.
- Attendance history and feedback trends.

HR dashboard metrics

- Total active employees, present/late counts, office vs WFH counts and pending requests.
- Department/date/status filters and employee attendance tables.
- HR's own attendance history and Admin feedback trends.

Admin dashboard metrics

- Active HR/employee counts, company-wide attendance trends, HR attendance and WFH statistics.
- Pending HR WFH requests, QR display status, HR feedback trends, and audit activity.

Attendance percentage formula

Proposed formula: (qualifying present days + qualifying WFH days) ÷ eligible working days × 100. Define treatment of holidays, weekly offs, incomplete records, approved exceptions and pending requests. Use the same definition across every dashboard and report.

Chart accuracy

- Charts and summary cards must use the same date filters and eligibility definitions as their source records.
- Office attendance and WFH attendance are mutually exclusive for a daily record.
- Working duration is check-out minus check-in; unfinished records do not count as completed durations.
- Report timezone and date range should be visible to the user.

11. Security, privacy and auditability

Authentication and role security

- Use Supabase Auth sessions and server-side membership/role checks.
- Never accept a client-supplied role as authoritative.
- Keep service-role keys and signing secrets out of frontend/mobile bundles.
- Validate all user inputs and rate-limit signup, QR validation, passcode attempts and sensitive endpoints.

RLS policy principles

- Profiles: users can access their own profile and permitted fields.
- Companies/memberships: users see only their memberships; Admin manages company membership within authorization.
- Employees/attendance: enforce company scope and ownership; employees only see their own records.
- WFH: employees read their own requests; HR reviews employee requests; Admin reviews HR requests.
- Passcodes/QR sessions: no unrestricted listing or direct token reads; use privileged validation functions.
- Feedback: subjects read their own reviews; reviewers can create reviews only within their permitted scope.
- Audit logs: append-only to normal application roles; read access limited to authorized administrators.

Privacy and demo data

- Use clearly identified fictional sample employee data.
- Do not use real employee personal information in demo seed data.
- Store only the profile and attendance data needed for the project.
- Biometric unlock is device-local; do not collect biometric templates.
- Do not log plaintext passwords, passcodes, QR tokens, or sensitive authentication material.

Audit events

- Company and membership creation/role changes.
- Attendance corrections and policy changes.
- WFH approval/rejection and passcode generation/consumption.
- QR display pairing/deactivation.
- Feedback creation and authorized changes.

12. UI and navigation structure

Design direction

Use a modern, premium, responsive interface with navy, blue, cyan and violet accents; clear typography; consistent spacing; readable tables; subtle transitions; and selective glassmorphism. Loading, empty, success and error states must be implemented for every important flow.

Website routes

Area	Suggested routes
Public/company	/, /company/signup, /company/login, /login, /forgot-password, /reset-password
Admin	/admin/dashboard, /admin/hr, /admin/attendance, /admin/wfh-requests, /admin/policies, /admin/qr-displays, /admin/feedback, /admin/profile
HR	/hr/dashboard, /hr/employees, /hr/attendance, /hr/wfh-requests, /hr/policies, /hr/mark-attendance, /hr/my-attendance, /hr/feedback, /hr/my-feedback, /hr/profile
QR display	/display/:displayId

Employee app screens

- Welcome, signup/activation, login, forgot password, biometric unlock.
- Home dashboard, QR scanner, scan result, WFH request form, WFH passcode entry.
- Attendance history, feedback and charts, profile/settings, logout.

Reusable UI components

- StatCard, StatusBadge, DataTable, FilterBar, DateRangePicker, PageHeader.
- LoadingSkeleton, EmptyState, ErrorState, ConfirmDialog, Toast.
- AttendanceTimeline, QRCountdown, FeedbackRatingForm, FeedbackTrendChart.

13. Error handling and edge cases

Scenario	Expected behavior
Unknown company or unauthorized signup	Reject; do not create privileged membership
Employee not pre-created by HR	Block activation
Expired QR	Reject and request a fresh QR
Display deactivated	Stop issuing/accepting its QR sessions
Duplicate QR request/retry	Return a safe result without duplicate attendance
Check-out before check-in	Reject
Second check-in after checkout	Reject under one-session-per-day policy
Late or missed WFH cutoff	Reject new request unless a separate authorized exception policy exists
Rejected/pending WFH request	Do not issue usable passcode or mark WFH
Expired/used/wrong-user passcode	Reject and audit security-relevant failures appropriately
Office record exists; WFH attempted	Reject conflict; never overwrite
Network failure during scan	Show retryable error; server idempotency prevents duplicate records
Missing check-out	Show In progress/Incomplete; do not invent duration
Unauthorized attendance correction	Reject and do not mutate record
RLS denies a query	Show a clear access error without exposing other-company data
Chart has no matching data	Show an empty state, not misleading zero-derived conclusions

14. Acceptance criteria and testing

Use these acceptance criteria as the definition of done. Every scenario should be tested with both successful and rejected inputs.

ID	Acceptance criterion
AC-01	Company signup creates a company and initial Admin securely.
AC-02	Unauthorized HR/Admin or employee activation is rejected.
AC-03	Employee profile data comes from the matching pre-created company record.
AC-04	Biometric unlock protects an existing authenticated session.
AC-05	QR expires after 18 seconds; old tokens are rejected server-side.
AC-06	Valid office scans create one check-in and one check-out daily.
AC-07	Late status follows assigned shift and grace-period rules.
AC-08	Two-hour WFH cutoff applies to both HR and employees.
AC-09	HR approves employee WFH; Admin approves HR WFH.
AC-10	Approved WFH requires a valid single-use passcode.
AC-11	Conflicting office/WFH records cannot overwrite each other.
AC-12	Filters apply consistently to attendance tables and charts.
AC-13	HR and Admin feedback includes period, ratings and written comments.
AC-14	Subjects can view feedback; unauthorized users cannot change it.
AC-15	RLS blocks cross-company access and role escalation.
AC-16	Sensitive changes create audit entries.
AC-17	System works end-to-end with fictional demo data.
AC-18	Project report documents architecture, features, tests and limitations.

Recommended test layers

- Unit tests: deadline calculations, late rules, status transitions, feedback validation and passcode expiry.
- Database tests: constraints, transactions, RLS policies, cross-company access and single-use passcode race conditions.
- API/function tests: invalid/expired QR, unauthorized role, duplicate scans, WFH approval hierarchy and retries.
- End-to-end tests: company signup → Admin adds HR → HR adds employee → employee activates → QR check-in/check-out → dashboard refresh.
- WFH end-to-end test: request before cutoff → correct reviewer approves → passcode generated → WFH attendance marked → history/chart updated.

15. Implementation roadmap

Phase	Deliverables
Phase 1 — Backend foundation	Create Supabase schema and migrations, company/auth onboarding, memberships, roles, RLS policies and seed data.
Phase 2 — Admin website	Build company setup, HR management, global policies, HR WFH approvals, QR controls, HR attendance and feedback.
Phase 3 — HR website	Build employee management, employee WFH approvals, attendance filters, personal attendance and feedback reviews.
Phase 4 — Office QR display	Implement display pairing, 18-second QR rotation, token expiry, server validation and attendance integration.
Phase 5 — Integration and testing	Connect website flows to Supabase; test permissions, attendance, WFH, analytics, error cases and security.
Phase 6 — Employee mobile app	Build React Native app: activation, login, biometrics, QR scanner, WFH requests, history, feedback and statistics.
Phase 7 — Final deployment/report	Deploy website and mobile app, configure production settings, run end-to-end tests and finalize the project report.

Suggested repository structure

```
vorldx-attendance/ apps/ web/ # React + Vite + Tailwind employee-mobile/ # React Native + Expo supabase/
migrations/ # Versioned SQL schema and constraints functions/ register-company/ activate-member/
issue-office-qr/ process-office-attendance/ submit-wfh-request/ review-wfh-request/ generate-wfh-passcode/
mark-wfh-attendance/ admin-correct-attendance/ submit-feedback-review/ seed.sql # Fictional demo data docs/
requirements-specification.pdf test-plan.md architecture.md README.md
```

Build the data model and authorization first. A polished UI should be layered on top of working business rules, not used as a substitute for them.

16. Demonstration scenario and project report

Recommended live demonstration

- Register a fictional company; confirm its initial Admin membership is created.
- Admin invites an HR user; HR activates the account and logs in.
- HR configures employees and shifts, then adds a fictional employee.
- Employee activates the account, views profile data, logs in, and enables biometric unlock.
- Admin pairs the office QR display; the display requests a QR session and refreshes every 18 seconds.
- Employee scans a valid QR, checks in, then checks out; exact timestamps appear in history and Admin/HR reports.
- Employee submits a WFH request before the two-hour cutoff; HR approves and a single-use passcode is generated.
- Employee marks WFH attendance with the passcode; the record appears as WFH and cannot be reused.
- HR submits a feedback review with scores and comments; employee sees the review and charts.
- Admin submits feedback for HR; HR sees the review and trends.
- Show rejected cases: expired QR, unauthorized signup, late WFH request, wrong/used passcode and role escalation.

Required project report contents

- Project overview, goals, scope, and functional requirements.
- Features implemented in the website, mobile app, and QR display.
- Frontend/backend/database architecture and data flow.
- Database entities, relationships, constraints, and access policies.
- Demo verification, QR attendance, check-in/check-out, WFH, and feedback workflows.
- Fictional sample data and simulated services.
- Screenshots of key screens and end-to-end flows.
- Testing performed, issues found, limitations, and features not completed.

17. Decisions and assumptions register

Confirmed by project owner

- Company signup automatically creates the initial Admin.
- HR approves employee WFH requests; Admin approves HR WFH requests.
- Approved WFH requires a single-use passcode to mark attendance.
- Admin controls global policies; HR views them.
- One check-in and one check-out per employee per day.
- Feedback includes ratings out of 10 and written comments.
- QR refresh interval is 18 seconds.
- Tech stack is React website, React Native mobile app, Supabase backend.
- WFH requests close two hours before the assigned shift; exact timestamps and filterable attendance are required.

Proposed implementation details to preserve consistency

- Company signup should create company and initial Admin membership through a trusted, failure-safe operation; email confirmation may be required before access.
- HR/employee signup should require a pre-created invitation or record; roles are assigned by server-side code.
- Use one daily attendance record with office/WFH as mutually exclusive attendance types.
- Use a single-use passcode bound to an approved WFH request, employee and date; store only its hash.
- Record actual WFH check-in/check-out timestamps rather than deriving them from approval time.
- Calculate attendance percentage from a documented denominator and keep attendance metrics separate from feedback scores.
- Use company timezone for shift deadlines and work dates; store timestamps as timestamptz.
- Use RLS, server-side authorization, audit logs, idempotency and transactions for sensitive operations.

Original brief exclusions retained

- No GPS/location tracking, Wi-Fi/Bluetooth proximity, biometric attendance verification, payroll, complex leave management, employee ranking, or advanced manager workflows in the initial scope.

End of specification

This document is a baseline for review and implementation planning. Features marked as proposed should be validated against company policy before production use. The final implementation should be judged against the acceptance criteria in Section 14.